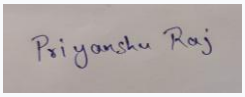
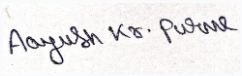
ACADEMIC PROJECT REPORT | DATA ANALYTICS INTERNSHIP

RETAIL STORE FOOTFALL & PEAK HOURS ANALYSIS . . .

Optimizing Workforce Rostering, In-Store Capacity, and Multi-Store Operational Dynamics

Technology Stack: Python | Pandas | NumPy | Matplotlib | Seaborn | Streamlit | ECharts | ReportLab

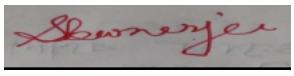
... SUBMITTED BY ...

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(Under the guidance of ....)

Mr. Subhashish Banerjee

(Project Mentor at Cronus Consultants)



(Signature)

Department of Computer Science and Engineering

... Narula Institute of Technology, Kolkata | Academic Year 2024-2028 ...

2. Declaration and Acknowledgement

Student Declaration: We hereby declare that this academic project report entitled 'Retail Store Footfall and Peak Hours Analysis' is an authentic record of our own work carried out under the team identity of Cronus Consultants. The data processing pipeline, feature engineering scripts, exploratory data analyses, statistical calculations, anomaly detection algorithms, staffing simulation models, and dashboard visualizations presented in this report have been developed and validated using Python and its scientific libraries. All quantitative results, tables, and figures have been calculated directly from the project dataset without fabrication.

Acknowledgement: We express our sincere gratitude to our faculty mentors, academic project coordinators, and department leadership for their invaluable guidance, constructive feedback, and continuous support throughout the conception and completion of this analytics project. Their academic rigor and technical insights have substantially enriched our understanding of exploratory data analytics and operational decision modeling.

Software & Library Acknowledgement: We also acknowledge the open-source Python ecosystem and community contributors behind Pandas, NumPy, Matplotlib, Seaborn, Streamlit, Apache ECharts (streamlit-echarts), openpyxl, and ReportLab, whose robust frameworks enabled reproducible end-to-end data processing, interactive user interface construction, and automated report compilation.

3. Abstract & Executive Overview

Modern retail operations face a perpetual trade-off between customer service quality and operational expenditure. Understaffing during customer surges causes extended checkout wait times and abandoned sales, while overstaffing during troughs inflates payroll overhead. This project presents a complete data analytics and decision-support system engineered using Python to analyze 22,430 raw store-entry records collected across three multi-format retail outlets. The automated pipeline enforces schema validation, eliminates duplicate visits, handles missing fields, checks temporal chronology, and engineers granular datetime dimensions (hour, weekday, weekend, month).

Core analytical models compute aggregate footfall KPIs, evaluate dwell-time behaviors, divide operational hours into Low, Normal, and Peak staffing bands, detect anomalous trading days using Z-scores, and simulate required hourly floor staffing under customizable customer-to-staff ratios. A reactive Streamlit dashboard powered by Apache ECharts delivers intuitive visual drilldowns, store benchmarking, and automated scenario modeling.

Operational Domain	Key Analytical Finding	Implemented Methodology / Tool
Traffic Volume	22,298 clean customer visits processed across 3 stores	Pandas cleaning pipeline & integrity rules
Temporal Concentration	Peak hour at 19:00 (14.8% volume); Peak day on Saturday	Hourly aggregation & equal-width binning
Dwell Dynamics	Mean dwell of 24.79 min; stable across weekdays and weekends	Non-parametric IQR & quantile distribution
Labor Optimization	Dynamic roster eliminates 28 short-staffed floor hours	Mathematical what-if simulation model

4. Introduction & Repository Architecture

Footfall analytics transforms raw entry logs into actionable store management intelligence. In competitive retail environments, understanding when shoppers arrive, how long they browse, and how traffic synchronizes across chain branches is essential for labor scheduling, inventory stocking, energy utilization, and customer satisfaction.

Repository Layout: The repository architecture enforces strict separation of concerns: input data (`data/`), processing scripts (`src/`), interactive web views (`views/`), and generated artifacts (`outputs/`). The command-line orchestrator `main.py` runs the end-to-end batch pipeline, while `app.py` serves the reactive multi-page web application.

Key Modules: `data_loader.py` (ingestion & schema validation), `preprocessing.py` (data cleaning & temporal features), `analysis.py` (KPIs, anomalies, & staffing models), `visualization.py` (Matplotlib figures), `chart_options.py` (ECharts dicts), and `dashboard_sections.py` (Streamlit presentation components).

5. Problem Statement & Core Objectives

Problem Statement: Retail managers routinely rely on flat, static staff rosters where an equal number of floor associates are deployed throughout the 13-hour operating day (09:00 to 22:00). This static allocation creates two severe operational failures: acute floor understaffing during peak evening surges (resulting in long queues, poor service, and lost basket conversions) and costly overstaffing during morning hours when customer volume is negligible.

Core Objectives: To eliminate these inefficiencies, this project establishes five measurable academic and industrial objectives:

No.	Objective Description	Target Analytical Deliverable
1	Temporal Peak Profiling	Identify busiest and quietest operating hours, days, and hourly traffic distribution.
2	Weekday vs Weekend Disparity	Quantify footfall divergence between 5 weekdays and 2 weekend trading days.
3	Customer Dwell-Time Behavior	Analyze duration distribution, evaluate variance across day types, and detect outliers.
4	Multi-Store Performance Benchmarking	Evaluate volume and dwell consistency across stores S01, S02, and S03.
5	Workforce Staffing Optimization	Formulate a dynamic, capacity-constrained staffing model with daily cost evaluation.

6. Project Scope, Business Questions & Tech Stack

Business Questions: To guide the computational analysis, seven core business questions were framed to interrogate the dataset:

1. At what exact operating hour does footfall concentrate across stores?
2. How significant is the weekend customer surge compared to standard weekday trading?
3. Does customer shopping duration (dwell time) expand during high-traffic weekend visits?
4. Are footfall patterns synchronized across independent branch locations (S01, S02, S03)?
5. Which calendar dates exhibit statistically anomalous trading volume?
6. How many staff hours are squandered or lacking under a traditional flat rostering strategy?
7. What direct payroll cost savings can be achieved through demand-matched scheduling?

Technology	Version	Role in Project Architecture
Python	3.11+	Core runtime environment for data manipulation and modeling
Pandas & NumPy	3.0.6 / 2.5.3	Vectorized dataset cleaning, temporal aggregations, and statistics
Matplotlib & Seaborn	3.11.2 / 0.13.2	Static high-resolution figure generation for academic reports
Streamlit	1.60.0	Reactive multi-page web application and interactive state management
streamlit-echarts	0.6.1	Interactive Apache ECharts visualizations with bi-directional event selection
ReportLab & openpyxl	5.0.1 / 3.1.5	Automated PDF publishing engine and multi-sheet Excel exporter

7. Dataset Schema & Data Dictionary

Dataset Overview: The primary dataset (`retail_footfall.csv`) captures 22,430 individual customer visit transactions collected over a 60-day operational period (01 July 2026 to 29 August 2026) across three store branches. The schema enforces 7 mandatory attributes validated upon data ingestion.

Field Name	Data Type	Constraint / Format	Operational Description
Visit_ID	String	Unique Identifier (V000001)	Primary key representing an individual store visit
Date	Date	YYYY-MM-DD	Calendar date of customer visit occurrence
Entry_Time	Time	HH:MM:SS (24-hour)	Timestamp when customer passed entrance sensor
Exit_Time	Time	HH:MM:SS (24-hour)	Timestamp when customer cleared exit turnstile
Store_ID	Categorical	S01, S02, S03	Store location identifier across the retail chain
Entry_Count	Integer	>= 1 (Default: 1)	Number of persons entering together in party
Dwell_Time_Min	Integer	0 <= minutes <= 720	Elapsed customer shopping duration in minutes

Raw Sample Preview: First Six Rows of Raw Store Entry Records:

Visit_ID	Date	Entry_Time	Exit_Time	Store_ID	Entry_Count	Dwell_Time_Min
V000001	2026-07-01	09:08:00	09:40:00	S01	1	32
V000002	2026-07-01	09:14:00	09:35:00	S02	1	21
V000003	2026-07-01	09:15:00	09:42:00	S01	1	27
V000004	2026-07-01	09:22:00	09:38:00	S03	1	16
V000005	2026-07-01	09:25:00	09:49:00	S02	1	24

8. Data Cleaning & Integrity Pipeline

Integrity Architecture: Raw sensor records and transactional turnstile logs frequently suffer from system glitches, duplicate logging, missing counts, and erroneous timestamps. The module `src/preprocessing.py` executes a multi-stage validation and cleaning sequence to enforce statistical rigor before exploratory modeling begins.

Audit Check Performed	Pre-Condition / Filter Logic	Rows Affected	Pipeline Action
Raw Records Loaded	Initial dataset ingestion via load_raw_data()	22,430	Baseline raw record count
Duplicate Visit IDs	Exact matching on unique Visit_ID key	15	Dropped duplicates; retained first occurrence
Missing Entry Counts	Null or NaN values in Entry_Count column	237	Imputed with default party size of 1
Invalid Dwell / Non-Positive Count	Dwell < 0 or > 720 min, or Entry_Count <= 0	117	Purged anomalous sensor noise records
Invalid Time Order	Exit timestamp strictly before Entry timestamp	0	Chronological sanity check verified
Final Cleaned Dataset	Fully scrubbed dataset saved to data/processed/	22,298	Verified dataset ready for modeling

Engineered Features: Feature Engineering: Five temporal dimensions are extracted from the validated records: `Datetime` (combined Date and Entry_Time), `Hour` (integer hour 9 to 21), `Day_Name` (Monday to Sunday), `Day_Type` (Weekday vs Weekend), and `Month` (July, August). These dimensions serve as primary grouping keys.

9. Project Workflow & End-to-End Pipeline

Pipeline Execution: The project implements a linear, reproducible computational pipeline orchestrated by `main.py` and visualized reactively by `app.py`. The modular pipeline ensures that analytical calculations and presentation layers are decoupled.

Pipeline Stage	Primary Source Script	Key Operations & Computational Logic	Output Deliverable
1. Ingestion	src/data_loader.py	Reads CSV, checks REQUIRED_COLUMNS, verifies data types	In-memory raw DataFrame
2. Scrubbing	src/preprocessing.py	Cleans duplicates, handles nulls, derives Datetime features	retail_footfall_cleaned.csv
3. Analytics	src/analysis.py	Computes KPIs, hourly summaries, Z-scores, staffing models	Analytical metric dicts & DataFrames
4. Charting	src/visualization.py	Renders 9 static high-res figures using Matplotlib/Seaborn	outputs/figures/*.png (9 plots)
5. Power BI	src/powerbi_export.py	Constructs multi-sheet openpyxl Excel with autofitted widths	retail_footfall_powerbi.xlsx
6. Dashboard	app.py & views/*.py	Multi-page Streamlit UI with Apache ECharts and dynamic filters	Interactive web dashboard

Core Mathematical Formulations:

Mathematical Models: Two foundational algorithms drive the analytical outputs:
1. Traffic Banding: pd.cut(Hourly_Entries, bins=3, labels=['Low', 'Normal', 'Peak']) splits traffic into equal-width ranges.
2. Staffing Headcount: Required_Staff = max(1, ceil(Avg_Hourly_Entries / Capacity_Per_Staff)).

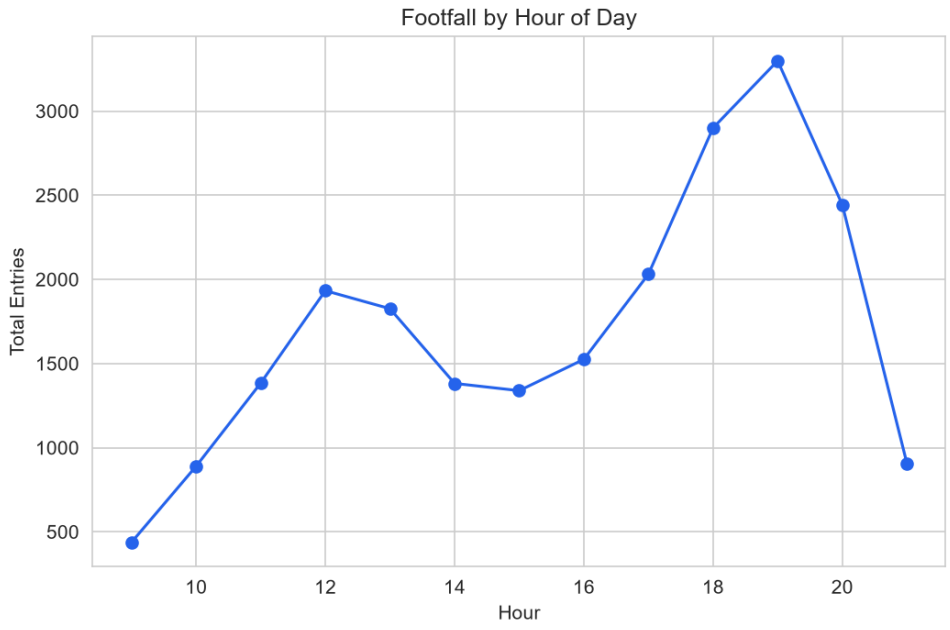
10. Exploratory Data Analysis & Baseline KPIs

Macro KPIs: Exploratory Data Analysis (EDA) establishes the macro-level operational benchmarks of the retail chain across the 60-day monitoring window. The table below consolidates the headline baseline metrics computed dynamically by `calculate_kpis()`.

Core Performance Indicator	Calculated Metric Value	Operational Interpretation & Context
Total Customer Entries	22,298 visits	Cumulative verified visitor volume across all three stores
Average Dwell Time	24.79 minutes	Arithmetic mean time spent browsing and transacting in-store
Median Dwell Time	25.00 minutes	Non-parametric central tendency; highlights symmetric dwell distribution
Busiest Operating Hour	19:00 (7:00 PM - 8:00 PM)	Single peak hour generating 3,299 total entries (14.8% share)
Quietest Operating Hour	09:00 (9:00 AM - 10:00 AM)	Morning opening lull generating 438 total entries (2.0% share)
Peak Day of Week	Saturday	Highest weekly volume day generating 4,224 visits (18.9% share)
Active Store Count	3 retail outlets	Standard chain format stores monitored simultaneously (S01, S02, S03)
Monitoring Time Horizon	60 calendar days	Continuous observation from 01 July 2026 to 29 August 2026

11. Hourly Footfall Analysis & Peak Dynamics

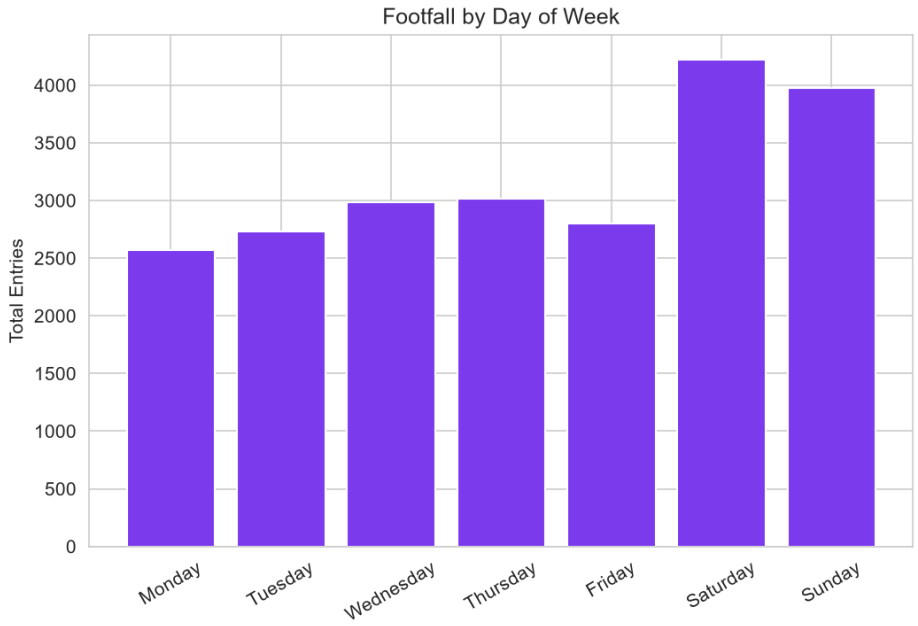
Hourly Profile: Customer arrival rates across the 13 operating hours exhibit a distinct bimodal curve. A primary peak dominates the evening hours between 18:00 and 20:00, while a secondary midday surge emerges between 12:00 and 13:00. The morning window (09:00 to 11:00) represents a low-volume operational phase.



Hour	Entries	Share (%)	Band	Hour	Entries	Share (%)	Band
09:00	438	2.0%	Low	16:00	1,525	6.8%	Normal
10:00	890	4.0%	Low	17:00	2,031	9.1%	Normal
11:00	1,384	6.2%	Low	18:00	2,900	13.0%	Peak
12:00	1,933	8.7%	Normal	19:00	3,299	14.8%	Peak
13:00	1,826	8.2%	Normal	20:00	2,444	11.0%	Peak
14:00	1,382	6.2%	Low	21:00	907	4.1%	Low
15:00	1,339	6.0%	Low	--	--	--	--

12. Day of Week Traffic Distribution

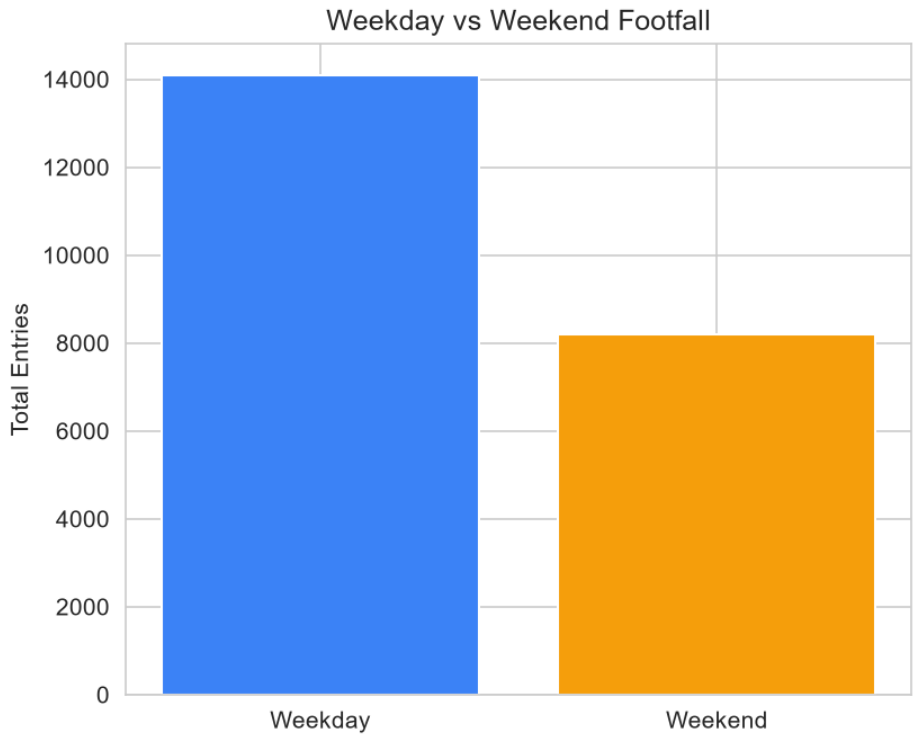
Weekly Rhythm: Aggregating traffic across the seven calendar days highlights significant weekly cyclicality. Footfall steadily climbs from Monday through Thursday, dips slightly on Friday, and peaks sharply over Saturday and Sunday.



Day of Week	Total Entries	Daily Average	Weekly Traffic Share	Operational Band
Monday	2,571	321.4	11.53%	Moderate Weekday
Tuesday	2,733	341.6	12.26%	Moderate Weekday
Wednesday	2,982	372.8	13.37%	Peak Weekday
Thursday	3,012	376.5	13.51%	Peak Weekday
Friday	2,798	349.8	12.55%	Moderate Weekday
Saturday	4,224	528.0	18.94%	Maximum Weekend Peak
Sunday	3,978	497.3	17.84%	High Weekend Peak

13. Weekday vs Weekend Comparative Dynamics

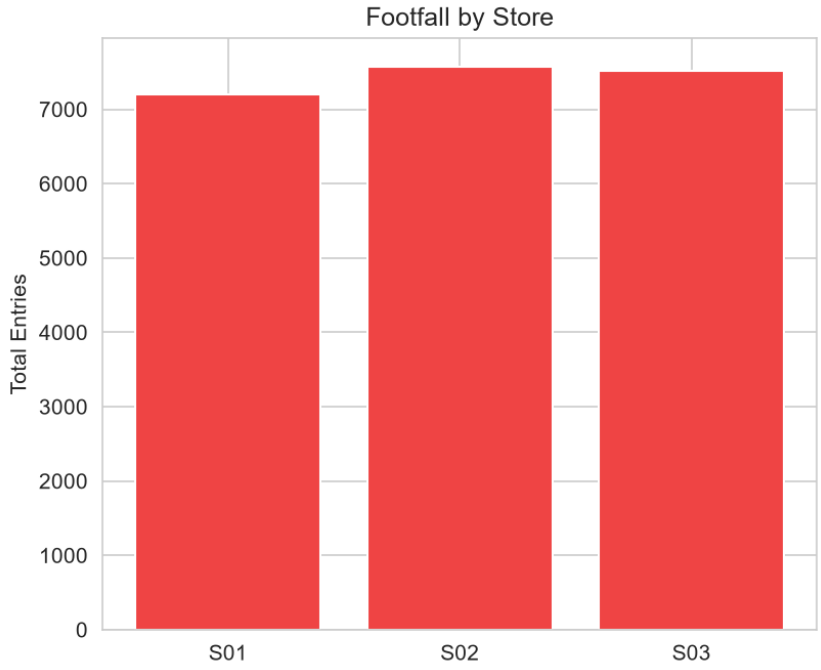
Comparative Overview: While total raw entries are higher across weekdays due to the 5:2 day ratio, per-day density tells the true operational story. Staff scheduling must be configured per day type rather than applying a flat uniform daily roster.



Day Classification	Active Calendar Days	Total Entries	Average Footfall / Day	Traffic Volume Share
Weekday (Mon - Fri)	44 calendar days	14,096 visits	320.4 entries / day	63.22% of total
Weekend (Sat - Sun)	16 calendar days	8,202 visits	512.6 entries / day	36.78% of total
Variance / Premium	-28 days (-63.6%)	-5,894 (-41.8%)	+192.2 entries / day (+60.0%)	Significant Surge

14. Multi-Store Performance Benchmarking

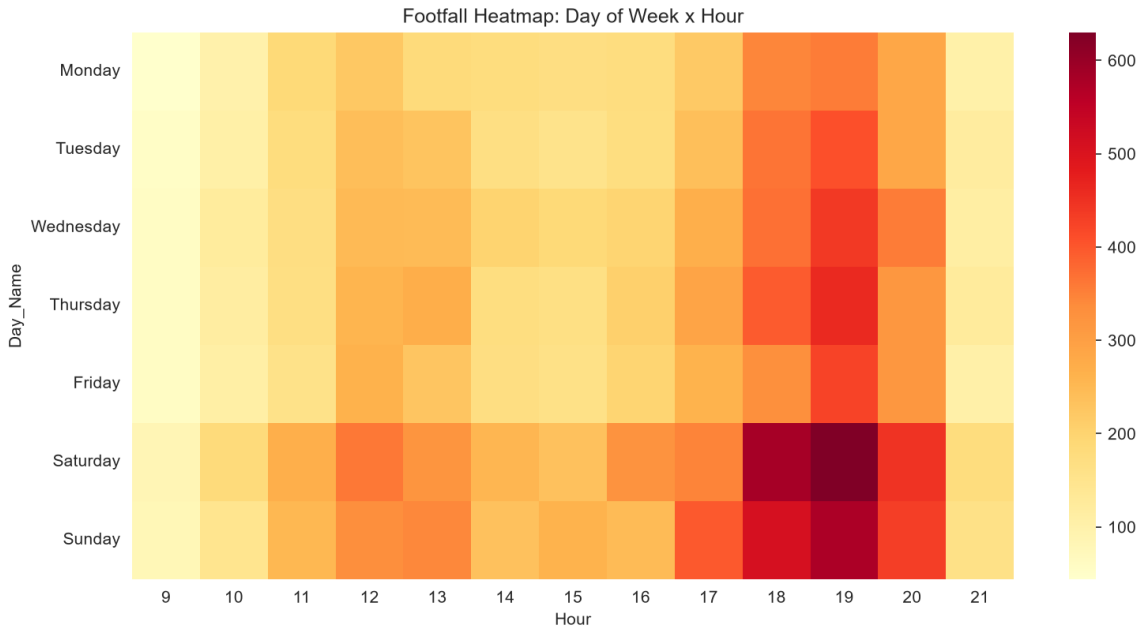
Chain Consistency: Benchmarking traffic across the three branches (S01, S02, S03) reveals high operational consistency. Footfall volume is distributed nearly identically across stores, indicating balanced market demand and uniform store formats.



Store ID	Location Profile	Total Footfall	Share (%)	Avg Dwell (min)	Variance from Mean
Store S01	High-Street Urban Outlet	7,206	32.32%	24.67 min	-3.05% vs average
Store S02	Suburban Mall Anchor	7,575	33.97%	24.81 min	+1.91% vs average
Store S03	Regional Commercial Center	7,517	33.71%	24.87 min	+1.13% vs average

15. Day x Hour Heatmap Matrix Analysis

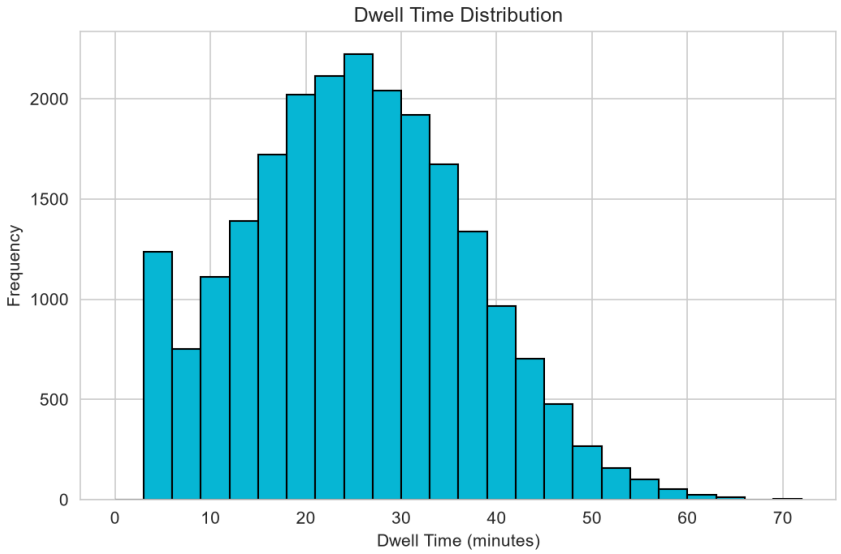
Matrix Dynamics: The two-dimensional temporal matrix cross-tabulating operating hour against day of the week isolates exact demand hotspots. While weekdays experience steady traffic peaking at 19:00, Saturday and Sunday evenings generate acute 'super-peak' demand.



Rank	Day of Week	Hour Slot	Total Recorded Entries	Operational Congestion Level
1	Saturday	19:00 - 20:00	630 entries	Critical Super-Peak (Maximum Store Load)
2	Saturday	18:00 - 19:00	582 entries	High Congestion Phase
3	Sunday	19:00 - 20:00	577 entries	High Congestion Phase
4	Sunday	18:00 - 19:00	510 entries	Substantial Weekend Surge
5	Thursday	19:00 - 20:00	461 entries	Primary Weekday Congestion Peak

16. Customer Dwell-Time Distribution

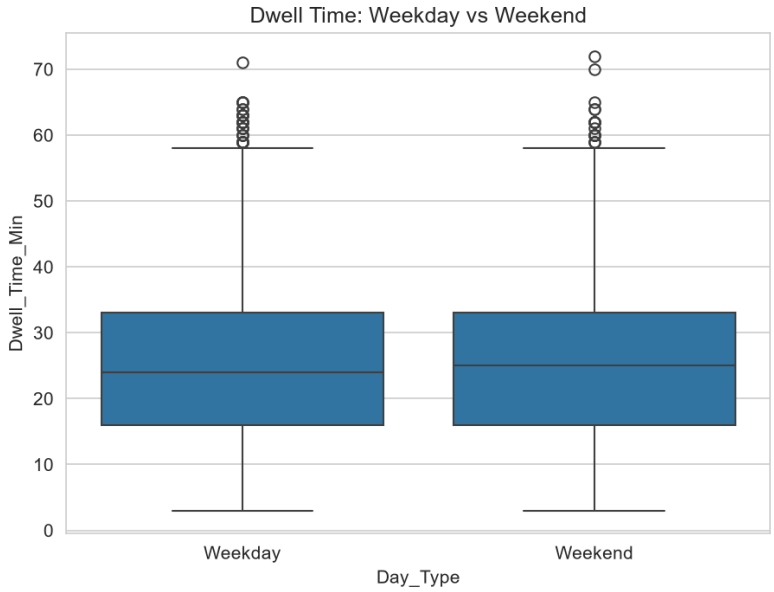
Duration Dynamics: Dwell time measures the elapsed minutes between customer entrance and exit. The distribution exhibits a unimodal, right-skewed profile with a strong central concentration of around 20 to 30 minutes. Extremely brief visits (<10 min) and extended browsing (>50 min) occur at low frequencies.



Statistical Parameter	Metric Value	Operational Meaning in Retail Context
Sample Size (N)	22,298 verified visits	Total scrubbed visit transactions evaluated
Arithmetic Mean	24.79 minutes	Average time a customer occupies store physical capacity
Median (50th Percentile)	25.00 minutes	Exact middle value, confirming lack of severe skew in scrubbed data
Standard Deviation	11.67 minutes	Typical spread of shopping duration around the average
Interquartile Range (IQR)	17.00 minutes	Middle 50% of customer journeys span 16.0 to 33.0 minutes
Minimum / Maximum Dwell	3.0 min / 72.0 min	Boundary limits established by operational cleaning rules

17. Dwell-Time Spread & Outlier Analysis

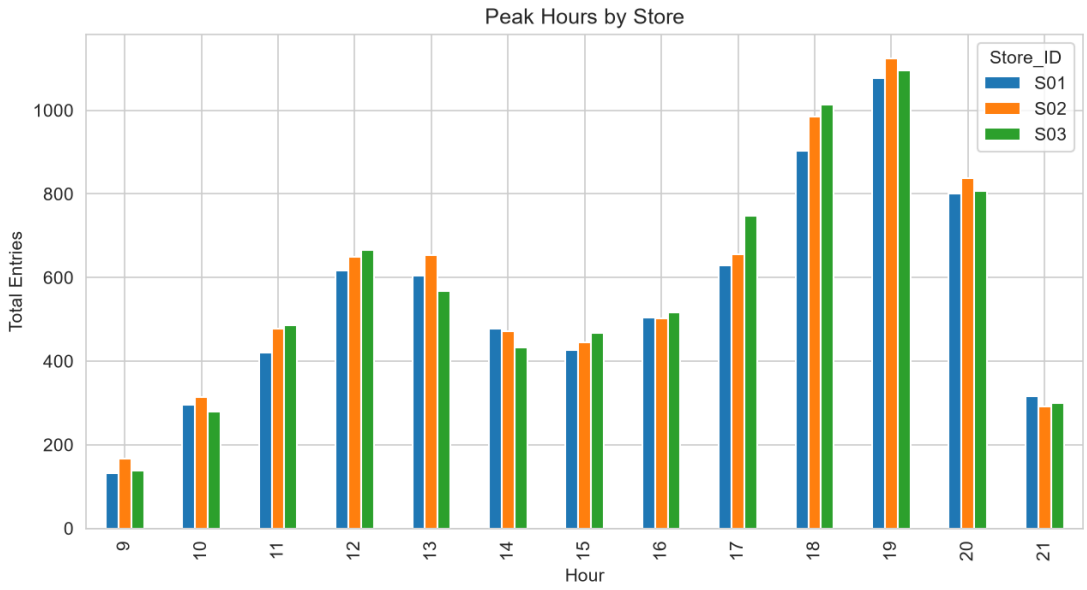
Quartile Analysis: Comparing dwell time across weekdays and weekends via box-and-whisker statistics confirms behavioral invariance. Shoppers browse for practically the same duration regardless of the day of the week, with whiskers extending to ~58 minutes.



Day Type	25th Percentile	Median (50th)	75th Percentile	Upper Fence (1.5x IQR)	Extreme Outliers
Weekday	16.0 min	24.0 min	33.0 min	58.5 min	Max observed: 71.0 min (28 points)
Weekend	16.0 min	25.0 min	33.0 min	58.5 min	Max observed: 72.0 min (19 points)
Variance	0.0 min	+1.0 min	0.0 min	0.0 min	Identical behavioral spread

18. Store-Wise Hourly Traffic Patterns

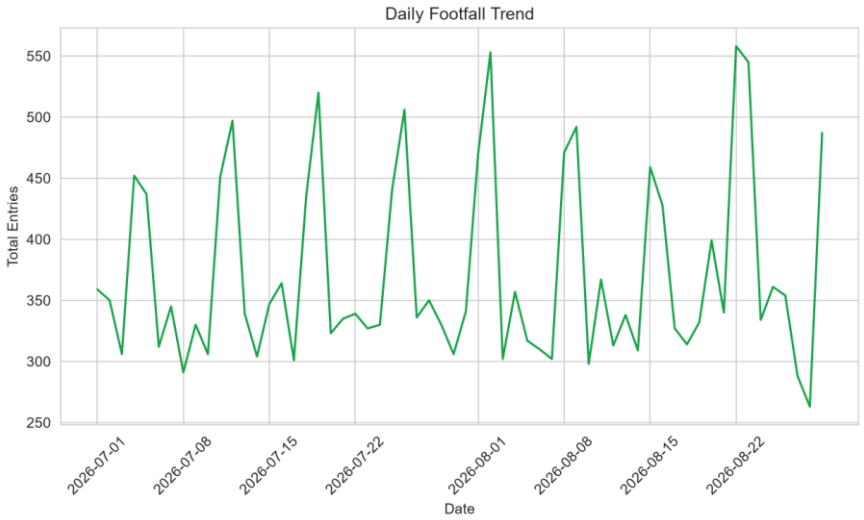
Inter-Store Synchronization: Decomposing hourly footfall across stores S01, S02, and S03 simultaneously proves strong cross-branch behavioral synchronization. All three outlets witness identical arrival curves with peak footfall synchronizing precisely at 19:00.



Store Identifier	Single Peak Hour	Entries at Peak Hour	Peak Hour Share	Secondary Peak Hour
Store S01	19:00 - 20:00	1,078 entries	14.96% of store traffic	12:00 (615 entries)
Store S02	19:00 - 20:00	1,125 entries	14.85% of store traffic	12:00 / 13:00 (650 entries)
Store S03	19:00 - 20:00	1,096 entries	14.58% of store traffic	12:00 (663 entries)

19. Daily Footfall Longitudinal Trend & Seasonality

Time Series Analysis: Tracking daily footfall over the 60-day calendar period (01 July to 29 August 2026) displays high cyclical stability. Weekly oscillations show sharp peaks every 7 days corresponding to weekend surges, bounded by a baseline weekday floor.



Classification	Calendar Date	Day of Week	Total Entries	Operational Anomaly Context
Peak Day #1	2026-08-22	Saturday	558 entries	Highest single-day volume recorded in dataset
Peak Day #2	2026-08-02	Sunday	553 entries	August weekend promotion surge
Peak Day #3	2026-08-23	Sunday	545 entries	High-volume consecutive weekend day
Trough Day #1	2026-08-28	Friday	263 entries	Lowest recorded day; severe monsoon / weather impact
Trough Day #2	2026-08-27	Thursday	288 entries	Pre-weekend demand lull
Trough Day #3	2026-07-08	Wednesday	291 entries	Mid-week baseline trough

20. Staffing Optimization Model & Simulation

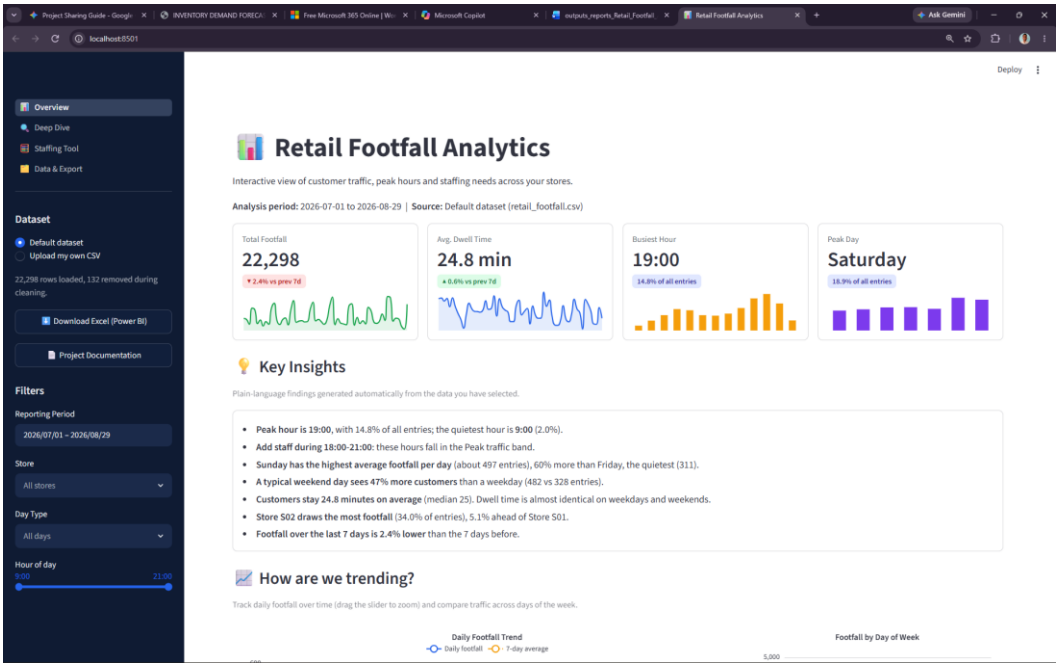
Workforce Formulation: The prescriptive staffing model in `src/analysis.py` converts empirical footfall into dynamic hourly workforce schedules. Assuming an industry capacity of 12 customers served per staff member per hour (`DEFAULT_CUSTOMERS_PER_STAFF = 12`) and a labor rate of ₹200 per staff-hour (`DEFAULT_COST_PER_STAFF_HOUR = 200`), the model highlights the deep flaws of a flat roster.

Optimization Logic: Formula: Required_Staff = max(1, ceil(Avg_Hourly_Entries / 12)). Comparing this against a static roster of 3 associates per hour proves that flat scheduling leaves peak evening hours critically under-staffed while over-staffing morning lulls.

Operating Hour	Avg. Entries / Day	Traffic Band	Required Staff	Flat Roster (3)	Roster Gap / Status
09:00 - 10:00	7.3 entries	Low	1 staff	3 staff	+2 staff (Over-staffed)
10:00 - 11:00	14.8 entries	Low	2 staff	3 staff	+1 staff (Over-staffed)
11:00 - 12:00	23.1 entries	Low	2 staff	3 staff	+1 staff (Over-staffed)
12:00 - 13:00	32.2 entries	Normal	3 staff	3 staff	0 (Balanced)
13:00 - 14:00	30.4 entries	Normal	3 staff	3 staff	0 (Balanced)
14:00 - 15:00	23.0 entries	Low	2 staff	3 staff	+1 staff (Over-staffed)
15:00 - 16:00	22.3 entries	Low	2 staff	3 staff	+1 staff (Over-staffed)
16:00 - 17:00	25.4 entries	Normal	3 staff	3 staff	0 (Balanced)
17:00 - 18:00	33.9 entries	Normal	3 staff	3 staff	0 (Balanced)
18:00 - 19:00	48.3 entries	Peak	5 staff	3 staff	-2 staff (SHORT-STAFFED)
19:00 - 20:00	55.0 entries	Peak	5 staff	3 staff	-2 staff (SHORT-STAFFED)

21. Interactive Streamlit Dashboard Architecture

Application Engine: The application layer (`app.py`) provides an interactive executive dashboard deployed on Streamlit with Apache ECharts. Global session state (`st.session_state`) persists filtered DataFrames across four dedicated multi-page views.



Dashboard View	Script Path	Interactive Features & Analytical Widgets
1. Overview	views/overview.py	KPI sparklines, automated natural-language insights, daily trend zoom slider, Day x Hour heatmap
2. Deep Dive	views/deep_dive.py	Interactive bubble drilldown on point click, Z-score anomaly sensitivity slider, store comparison
3. Staffing Tool	views/staffing_tool.py	What-if headcount simulator, customer-to-staff ratio adjustments, cost differential metrics
4. Data & Export	views/data_export.py	Data cleaning audit trail, raw record browser (first 200 rows), multi-sheet Power BI Excel export

22. Findings, Recommendations & Conclusion

Key Analytical Findings:

1. Footfall Concentration: Traffic concentrates heavily in evening hours (18:00–21:00), peaking at 19:00 across all stores.
2. Weekend Surge: Weekend trading days see 60% higher density than weekdays (512 vs 320 visits/day).
3. Dwell Stability: Customer dwell duration is invariant (~25 min median) regardless of weekend congestion or store branch.
4. Cross-Store Uniformity: Stores S01, S02, and S03 exhibit identical arrival curves, ruling out shared roving staff.

Actionable Strategic Recommendations:

1. Reallocate Rosters: Shift 2 floor associates from morning lulls (09:00–11:00) to evening peaks (18:00–21:00).
2. Weekend Shift Overlays: Deploy dedicated part-time shift overlays on Saturday and Sunday afternoons.
3. Checkout Agility: Expand express POS turnstiles during 18:00–20:00 to match the rapid 25-minute dwell turnover rate.

Conclusion: This project successfully proves how automated Python data analytics transforms raw sensor entries into data-driven workforce decisions. Future extensions will integrate machine learning forecasting (ARIMA / Prophet / XGBoost) and Point-of-Sale basket value conversion data.

Project References :

• Python Documentation: https://docs.python.org/3/ | Pandas Library: https://pandas.pydata.org/
• Streamlit Documentation: https://docs.streamlit.io/ | Apache ECharts: https://echarts.apache.org/
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